

Anatomically-Constrained LDMs for Rare Case Augmentation



SAPIENZA
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Context and objective:

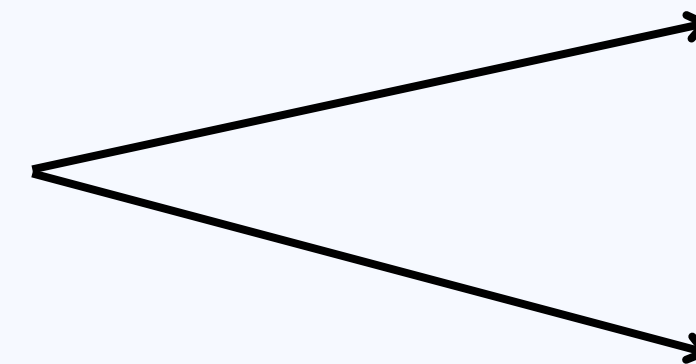
Periodontitis is a severe gum infection that destroys the soft tissue and bone supporting the teeth, representing one of the leading causes of tooth loss worldwide

Goal:

LDM



Anatomically-
Constrained
LDMs

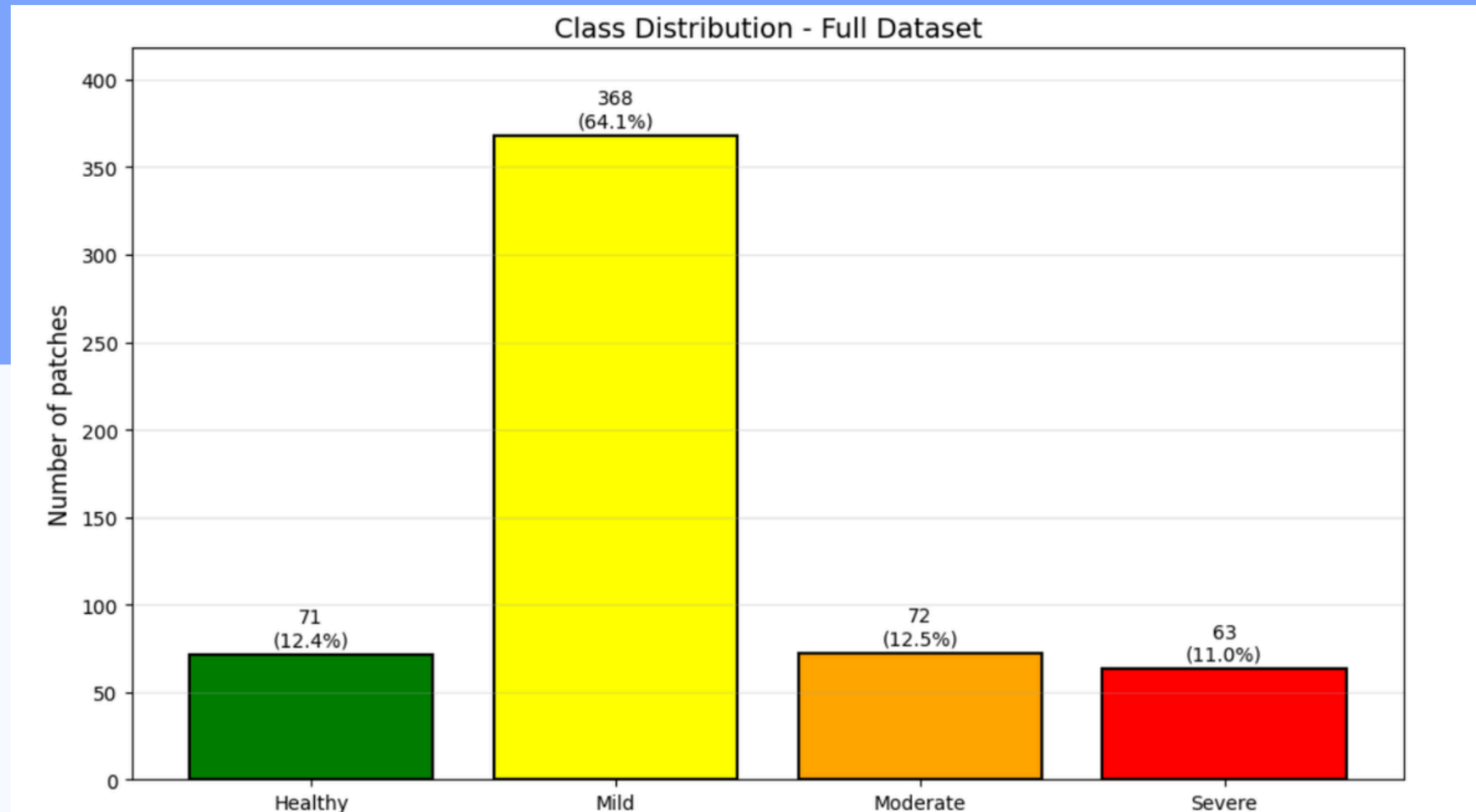


Downstream

Ablation study

Data organisation

Requirement : Keypoints in the image to allow us to constrain the LMD anatomically



Dataset already used in Periodontal Bone Loss Analysis via Keypoint Detection With Heuristic Post-Processing, Ryan Banks

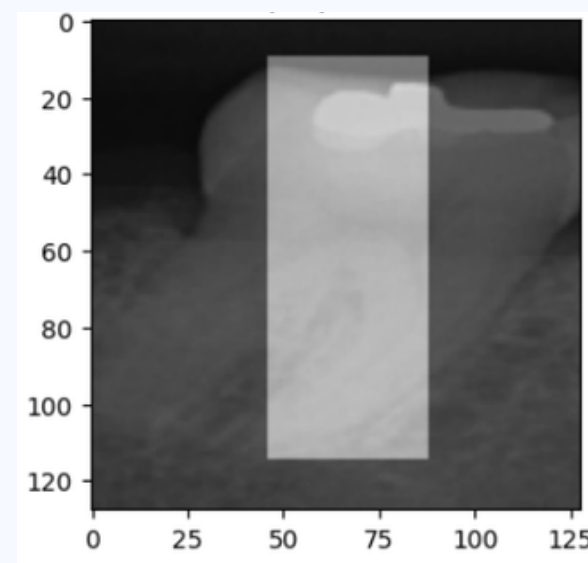
Our models:

LDM

- Standard LDM without spatial guidance
- Baseline for comparing the impact of anatomical constraints
- Generated using only random noise

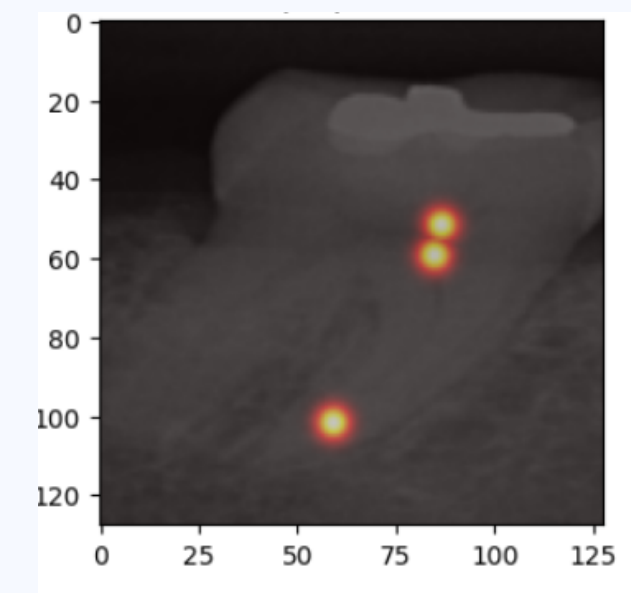
LDM with BBOX

- LDM guided by a simple binary mask
- Low spatial constraint (approximate region of the tooth)

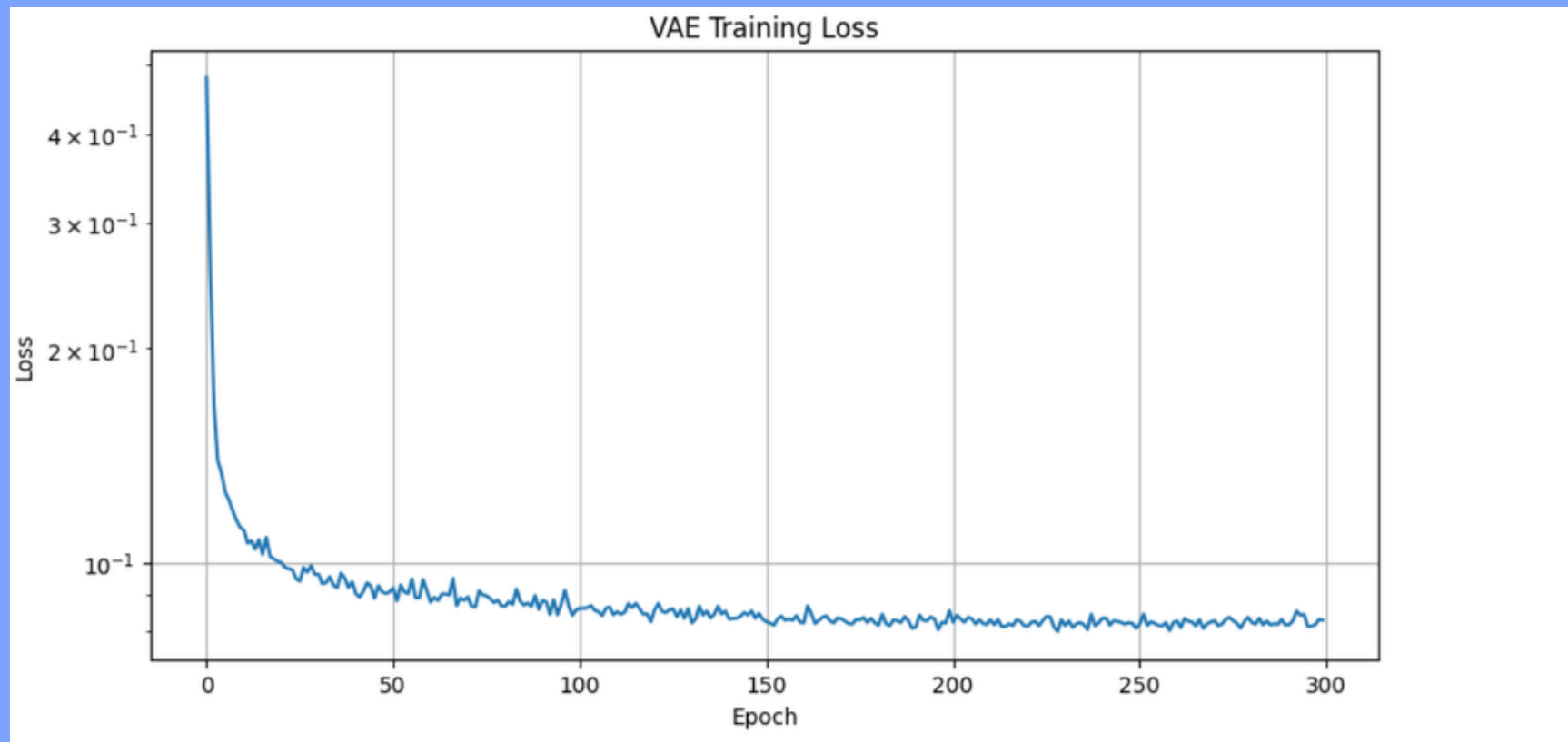


LDM with gaussian heatmap

- Guided by a continuous heatmap (0 to 1)
- Precise anatomical guidance (CEJ points, bone level, apex)



Evaluation:



Downstream

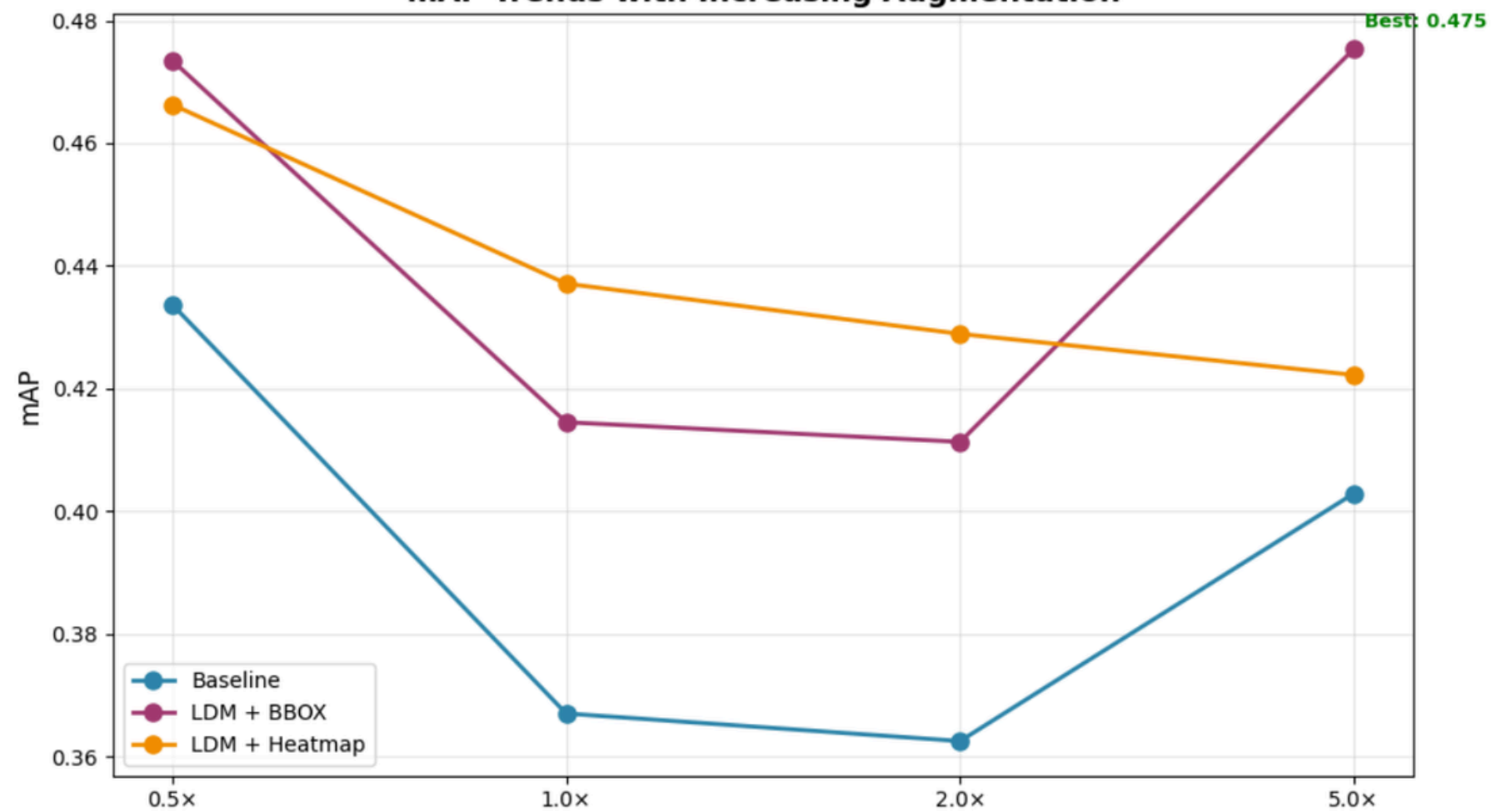
- Compare performance with and without synthetic data
- Metrics: mAP, AP (Severe)

Ablation Study

- Analyze the influence of the anatomical loss weight
- Identify the most effective design choices

Downstream:

mAP Trends with Increasing Augmentation

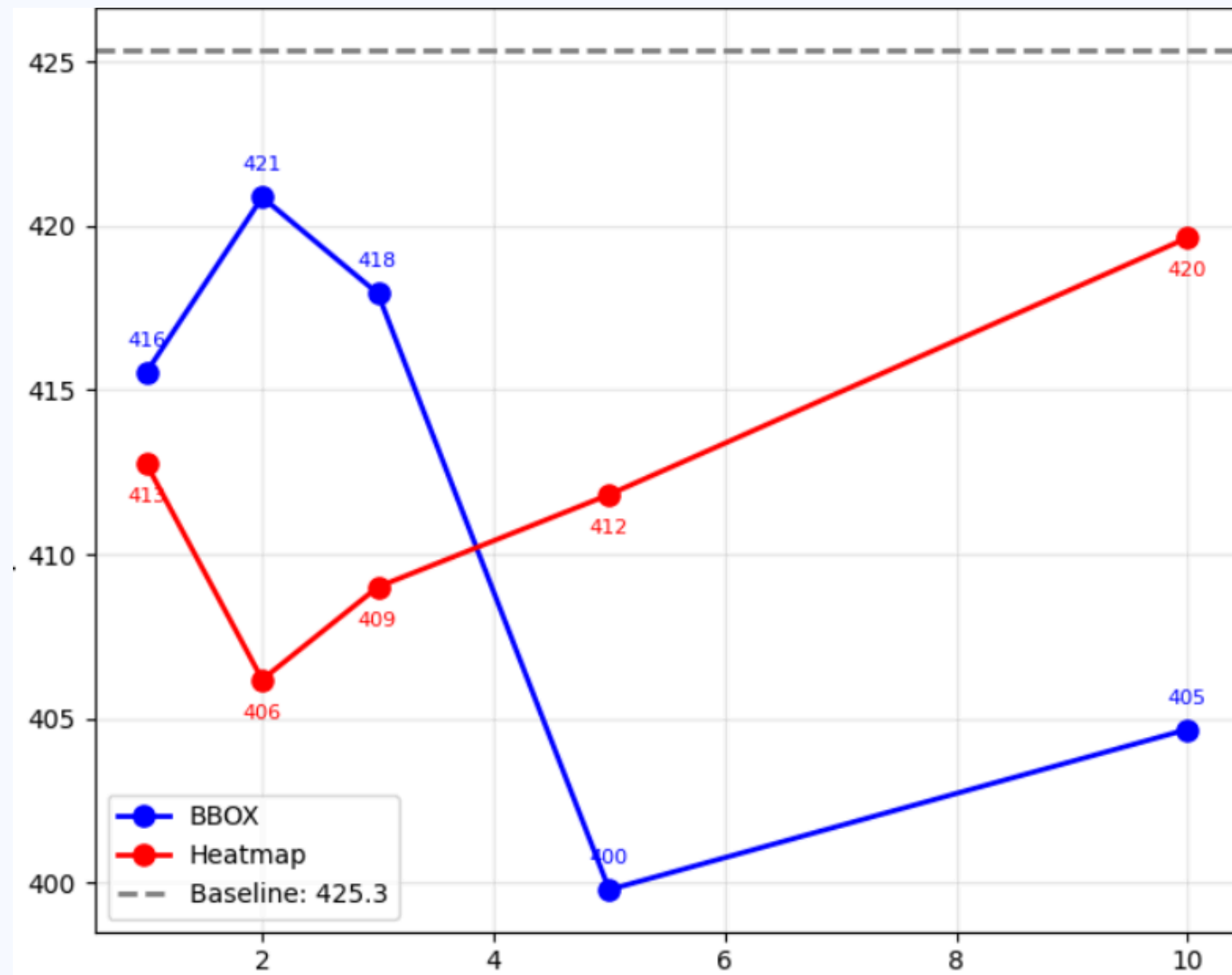


Severe Class Performance Comparison



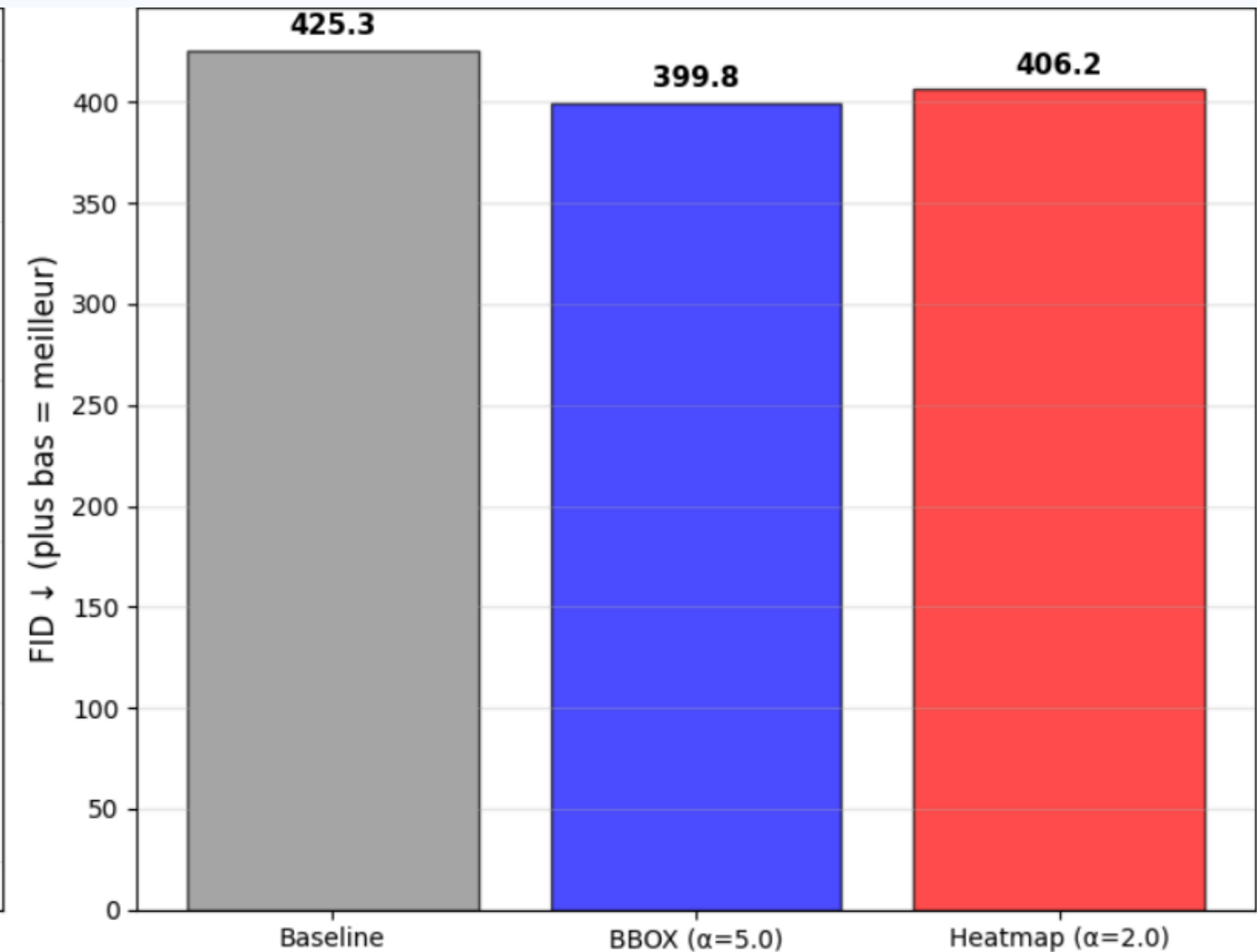
Ablation Study:

Impact of anatomic weight on FID



α anatomic weight

Best FID for each model



Conclusion